

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; gpt-5.4-nano

Abstract

This report evaluates `openai/gpt-5.4-nano` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Observed cost in all routing modes; Infron leads Throughput in all routing modes; OpenRouter leads E2E latency in all routing modes; OpenRouter leads Streaming TTFT in all routing modes.

Overall, Infron's cross-mode strengths are throughput, observed cost, while OpenRouter's cross-mode strengths are Streaming TTFT, E2E latency, cache reuse. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Inference Platform Impossible Quadrilateral

The five directions represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform–level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput Infron wins 4/4 Max advantage 2.61%, higher is better	Observed cost Infron wins 4/4 Max advantage 14.54%, lower is better	E2E latency OpenRouter wins 4/4 Max advantage 25.05%, lower is better	Streaming TTFT OpenRouter wins 4/4 Max advantage 11.30%, lower is better	Cache hit rate OpenRouter wins 3/4 Max advantage 3.90%, higher is better
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Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	Infron advantage 2.52%	Infron advantage 12.51%	OpenRouter advantage 19.78%	OpenRouter advantage 10.85%	OpenRouter advantage 0.40%
Price First price	Infron advantage 2.61%	Infron advantage 2.18%	OpenRouter advantage 15.58%	OpenRouter advantage 0.15%	OpenRouter advantage 3.67%
Latency First latency	Infron advantage 2.39%	Infron advantage 14.54%	OpenRouter advantage 22.52%	OpenRouter advantage 7.38%	Infron advantage 0.26%
TTFT First ttft	Infron advantage 2.37%	Infron advantage 0.25%	OpenRouter advantage 25.05%	OpenRouter advantage 11.30%	OpenRouter advantage 3.90%

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second <code>usage.prompt_tokens</code> deltas are limited to 50 tokens within each <code>sort/group/round</code> pair.	Methods and data quality
Cache reuse	Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in all routing modes	Overall metrics and provider drill-down
Performance	Infron leads Throughput in all routing modes; OpenRouter leads E2E latency in all routing modes; OpenRouter leads Streaming TTFT in all routing modes	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	OpenRouter	Infron	Infron	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	OpenRouter	Infron	Infron	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Latency First	Minimize full–response waiting time	Infron	Infron	Infron	OpenRouter	OpenRouter	Infron leads overall (3/5 metrics)
TTFT First	Minimize streaming first–token time	OpenRouter	Infron	Infron	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference–platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first–token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token–level cache hit rate for repeated stable prefixes.	Second–request cache–read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response–returned `usage.prompt_tokens` as the input–token control while allowing small cross–platform accounting variance up to 50 tokens.
- Extends prompt–caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

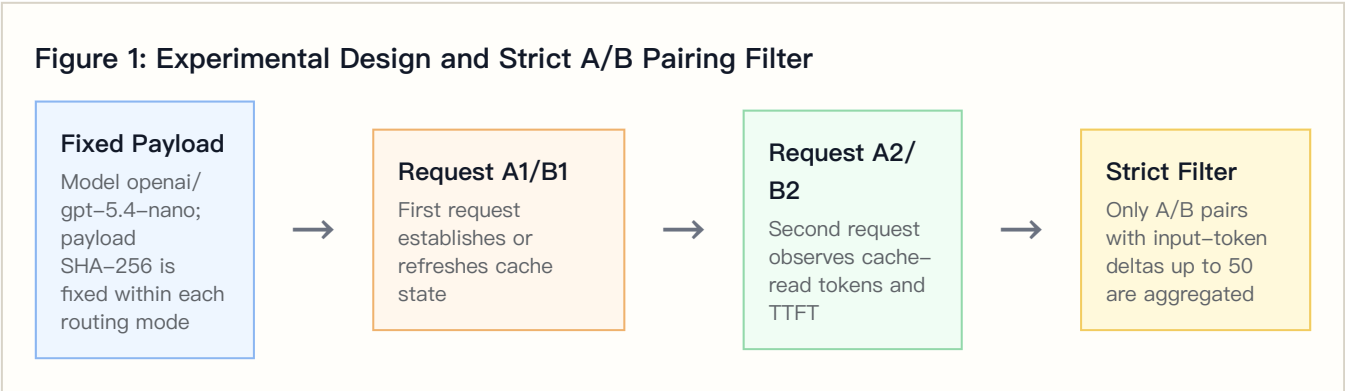
2. Experimental Design, Dataset, and Controls

2.1 Dataset Construction

The dataset is `business_representative`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with <code>cache_read_tokens > 0</code>	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	openai/gpt-5.4-nano
Provider model IDs	infron: openai/gpt-5.4-nano ; openrouter: openai/gpt-5.4-nano
Platforms	Infron and OpenRouter
API protocol	/v1/chat/completions
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	short ≈1500, medium ≈8000, long ≈32000
Excluded records	20

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	200	6262580	92.54%	\$0.21638200	18.24 tok/s	3196.16 ms	2422.16 ms
Throughput First	OpenRouter	200	6262580	92.91%	\$0.24345791	5.18 tok/s	2668.31 ms	2185.01 ms
Price First	Infron	194	5964592	93.81%	\$0.15337700	18.45 tok/s	3161.83 ms	2373.12 ms
Price First	OpenRouter	194	5964592	97.25%	\$0.15672743	5.11 tok/s	2735.66 ms	2369.49 ms
Latency First	Infron	198	6117118	95.38%	\$0.14883400	18.23 tok/s	3207.86 ms	2419.43 ms
Latency First	OpenRouter	198	6117118	95.12%	\$0.17046894	5.38 tok/s	2618.18 ms	2253.12 ms
TTFT First	Infron	198	6117122	93.46%	\$0.15877500	17.63 tok/s	3321.51 ms	2498.98 ms
TTFT First	OpenRouter	198	6117122	97.11%	\$0.15917730	5.23 tok/s	2656.24 ms	2245.33 ms

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	2955.57 ms	4934.77 ms	6023.18 ms	2227.26 ms	3945.49 ms	5730.02 ms
Throughput First	OpenRouter	2396.26 ms	4269.54 ms	10115.76 ms	2020.34 ms	3519.31 ms	4767.88 ms
Price First	Infron	3019.49 ms	4669.59 ms	6120.17 ms	2222.34 ms	3599.42 ms	4969.66 ms
Price First	OpenRouter	2463.58 ms	4238.64 ms	5744.71 ms	2156.99 ms	3669.98 ms	5449.70 ms
Latency First	Infron	3054.49 ms	4658.03 ms	5847.52 ms	2193.80 ms	3677.12 ms	4926.94 ms
Latency First	OpenRouter	2392.75 ms	4109.78 ms	6010.21 ms	2087.42 ms	3573.68 ms	5224.04 ms
TTFT First	Infron	3067.54 ms	5172.86 ms	7030.92 ms	2242.49 ms	4074.42 ms	6279.34 ms
TTFT First	OpenRouter	2432.86 ms	4209.05 ms	5721.74 ms	2112.12 ms	3730.59 ms	5219.15 ms

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	–1055.70 ms	–1452.44 ms to –655.74 ms	<0.001	200	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	–474.31 ms	–813.67 ms to –167.56 ms	0.0018	200	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	12.6346 tok/s	10.8967 tok/s to 14.4725 tok/s	<0.001	200	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	\$0.00013538	\$–0.00005966 to \$0.00031376	0.1690	200	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	24.34 pp	17.41 pp to 30.72 pp	<0.001	200	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	–852.35 ms	–1242.40 ms to –382.43 ms	<0.001	194	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	–7.27 ms	–346.64 ms to 447.53 ms	0.9788	194	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	12.1359 tok/s	10.3406 tok/s to 13.9050 tok/s	<0.001	194	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	\$0.00001727	\$–0.00009310 to \$0.00012059	0.7486	194	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	–1.27 pp	–4.24 pp to 1.76 pp	0.5349	194	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	–1179.35 ms	–1523.12 ms to –842.98 ms	<0.001	198	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	–332.61 ms	–682.64 ms to –20.97 ms	0.0310	198	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	12.1062 tok/s	10.3876 tok/s to 13.8536 tok/s	<0.001	198	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	\$0.00010927	\$–0.00002411 to \$0.00026074	0.1232	198	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	–2.30 pp	–5.20 pp to 0.59 pp	0.1025	198	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	–1330.55 ms	–1592.49 ms to –1087.56 ms	<0.001	198	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	–507.29 ms	–741.64 ms to –280.94 ms	<0.001	198	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	11.8163 tok/s	10.0745 tok/s to 13.5822 tok/s	<0.001	198	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	\$0.00000203	\$–0.00011899 to \$0.00010971	0.9763	198	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	–1.55 pp	–4.38 pp to 0.98 pp	0.2927	198	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	0	0.0000	0	0.00 ms	2422.16 ms	3196.16 ms
Throughput First	OpenRouter	0	0.0000	0	0.00 ms	2185.01 ms	2668.31 ms
Price First	Infron	0	0.0000	0	0.00 ms	2373.12 ms	3161.83 ms
Price First	OpenRouter	0	0.0000	0	0.00 ms	2369.49 ms	2735.66 ms
Latency First	Infron	0	0.0000	0	0.00 ms	2419.43 ms	3207.86 ms
Latency First	OpenRouter	0	0.0000	0	0.00 ms	2253.12 ms	2618.18 ms
TTFT First	Infron	0	0.0000	0	0.00 ms	2498.98 ms	3321.51 ms
TTFT First	OpenRouter	0	0.0000	0	0.00 ms	2245.33 ms	2656.24 ms

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions` ; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
chat_completions	/v1/chat/completions	Infron	800	1600	99.25%	100.00%	100.00%	100.00%	100.00%
chat_completions	/v1/chat/completions	OpenRouter	800	1600	100.00%	100.00%	100.00%	100.00%	100.00%

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

**Figure 3-E2:
Normalized
capability
contour**

Five metrics are normalized to 0-100, where higher is better.

**Figure 3-E3:
Cost-cache
efficiency plane**

X-axis is token cache hit rate; Y-axis is observed total cost.

**Figure 3-E1:
Routing-mode
core
metric
comparison**

Infron and OpenRouter are shown side by side by routing mode.

Figure 3–E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3–E5: Upstream provider distribution

Provider shares explain cache–domain and performance differences.

Routing–Mode Drill–Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

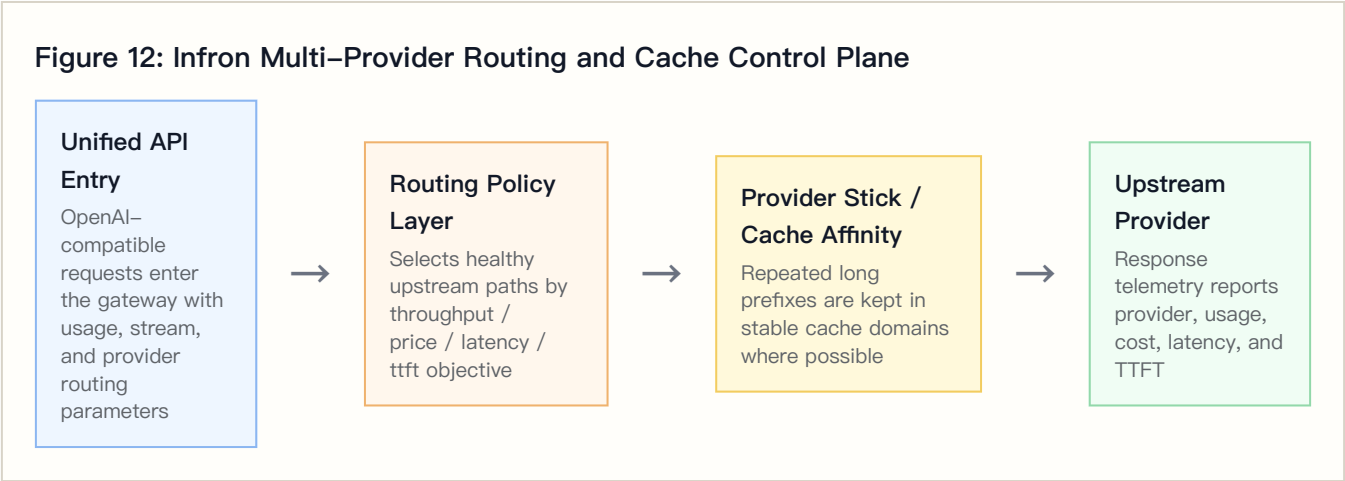
Figure 4–E3: Latency First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	400	400	azure 400
Throughput First	OpenRouter	400	400	OpenAI 400
Price First	Infron	388	388	azure 388

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Price First	OpenRouter	388	388	OpenAI 258, Azure 130
Latency First	Infron	396	396	azure 396
Latency First	OpenRouter	396	396	OpenAI 262, Azure 134
TTFT First	Infron	396	396	azure 396
TTFT First	OpenRouter	396	396	OpenAI 262, Azure 134

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/ second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens	Reasoning tokens
Throughput First	Infron	azure	400	100.00%	200/200	200	2422.16 ms	3196.16 ms	6262580	23320	0
Throughput First	OpenRouter	OpenAI	400	100.00%	200/200	200	2185.01 ms	2668.31 ms	6262580	5531	0
Price First	Infron	azure	388	100.00%	194/194	194	2373.12 ms	3161.83 ms	5964592	22628	0
Price First	OpenRouter	OpenAI	258	66.49%	129/129	129	2634.74 ms	2985.22 ms	5734596	3591	0
Price First	OpenRouter	Azure	130	33.51%	65/65	65	1843.05 ms	2240.37 ms	229996	1836	0
Latency First	Infron	azure	396	100.00%	198/198	198	2419.43 ms	3207.86 ms	6117118	23162	0
Latency First	OpenRouter	OpenAI	262	66.16%	131/131	131	2525.49 ms	2893.22 ms	5880054	3742	0
Latency First	OpenRouter	Azure	134	33.84%	67/67	67	1720.59 ms	2080.43 ms	237064	1840	0
TTFT First	Infron	azure	396	100.00%	198/198	198	2498.98 ms	3321.51 ms	6117122	23186	0
TTFT First	OpenRouter	OpenAI	262	66.16%	131/131	131	2478.41 ms	2859.22 ms	5880058	3671	0
TTFT First	OpenRouter	Azure	134	33.84%	67/67	67	1789.62 ms	2259.38 ms	237064	1835	0

7.1 Cache–Rate and Cost Divergence Drill–Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing–mode differences.

Routing mode	Cache-hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	−0.37 pp	0.89x	azure 100.00%	OpenAI 100.00%	+0	Cache and cost move in different directions; evaluate with speed metrics
Price First	−3.44 pp	0.98x	azure 100.00%	OpenAI 66.49%	+0	Cache and cost move in different directions; evaluate with speed metrics
Latency First	+0.25 pp	0.87x	azure 100.00%	OpenAI 66.16%	+0	Infron leads both cache and cost
TTFT First	−3.65 pp	1.00x	azure 100.00%	OpenAI 66.16%	+0	Cache and cost move in different directions; evaluate with speed metrics

8. Stratified Results: Prompt–Length Cache Performance

This section aggregates second-request cache-read tokens, token-level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt-length tier. Bold cells mark the advantaged side within each tier.

Prompt–Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
short	1500	Infron	266	470594	393216	83.56%	\$0.06510800	2650.17 ms	1828.76 ms
short	1500	OpenRouter	266	470594	297984	63.32%	\$0.09124303	2213.67 ms	1779.54 ms
medium	8000	Infron	268	2449832	2302720	93.99%	\$0.14227500	3086.51 ms	2314.54 ms
medium	8000	OpenRouter	268	2449832	2383360	97.29%	\$0.13474755	2558.15 ms	2193.08 ms
long	32000	Infron	256	9310280	8774912	94.25%	\$0.46998500	3958.26 ms	3171.55 ms
long	32000	OpenRouter	256	9310280	9007616	96.75%	\$0.50384100	3258.96 ms	2837.01 ms

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
short	Throughput First	80.34%	0.00%	Infron
short	Price First	86.82%	81.48%	Infron

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
short	Latency First	81.64%	86.82%	OpenRouter
short	TTFT First	85.53%	85.53%	tie
medium	Throughput First	92.17%	96.56%	OpenRouter
medium	Price First	92.17%	98.02%	OpenRouter
medium	Latency First	95.09%	98.02%	OpenRouter
medium	TTFT First	96.56%	96.56%	tie
long	Throughput First	93.24%	96.56%	OpenRouter
long	Price First	94.61%	97.84%	OpenRouter
long	Latency First	96.15%	94.79%	Infron
long	TTFT First	93.05%	97.84%	OpenRouter

9. Stratified Results: Group–Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	83.84%	\$0.09584800	4932.61 ms	3955.95 ms
Infron	2	50	50	98.38%	\$0.03526900	4944.49 ms	3919.29 ms
Infron	3	50	50	93.82%	\$0.04386800	4589.81 ms	3776.66 ms
Infron	4	50	50	93.81%	\$0.04139700	5520.54 ms	4071.87 ms
OpenRouter	1	50	50	94.04%	\$0.08094779	4105.60 ms	3027.83 ms
OpenRouter	2	50	50	90.59%	\$0.06182644	4411.93 ms	3585.49 ms
OpenRouter	3	50	50	94.17%	\$0.05021102	4227.23 ms	3552.50 ms
OpenRouter	4	50	50	92.88%	\$0.05047266	3898.83 ms	2958.03 ms

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	97.35%	\$0.03795100	4616.45 ms	3606.14 ms
Infron	2	49	49	91.49%	\$0.04011900	4681.38 ms	3591.19 ms
Infron	3	46	46	88.99%	\$0.04370300	4827.88 ms	3755.10 ms
Infron	4	49	49	97.25%	\$0.03160400	4473.22 ms	3460.36 ms
OpenRouter	1	50	50	97.45%	\$0.04113467	4252.44 ms	3632.78 ms
OpenRouter	2	49	49	96.87%	\$0.04038242	4273.23 ms	3956.76 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
OpenRouter	3	46	46	97.26%	\$0.03747866	3881.69 ms	3045.69 ms
OpenRouter	4	49	49	97.43%	\$0.03773168	4755.65 ms	4033.56 ms

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	87.74%	\$0.04320100	4679.88 ms	3955.10 ms
Infron	2	49	49	96.97%	\$0.03291100	4204.07 ms	3528.29 ms
Infron	3	49	49	98.32%	\$0.04122100	4629.47 ms	3420.94 ms
Infron	4	50	50	98.51%	\$0.03150100	4804.74 ms	3981.56 ms
OpenRouter	1	50	50	97.45%	\$0.03949454	4043.33 ms	3374.09 ms
OpenRouter	2	49	49	92.82%	\$0.04576162	3753.61 ms	3500.40 ms
OpenRouter	3	49	49	97.44%	\$0.03931312	4775.16 ms	4261.07 ms
OpenRouter	4	50	50	92.81%	\$0.04589966	4133.43 ms	3240.43 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	93.81%	\$0.03661500	5227.76 ms	4093.91 ms
Infron	2	49	49	98.54%	\$0.03128400	5545.56 ms	4657.69 ms
Infron	3	49	49	87.61%	\$0.04835800	4627.42 ms	3696.16 ms
Infron	4	50	50	93.81%	\$0.04251800	4737.60 ms	3677.95 ms
OpenRouter	1	50	50	97.45%	\$0.03951579	3848.37 ms	3435.28 ms
OpenRouter	2	49	49	97.27%	\$0.03954508	4207.71 ms	3590.66 ms
OpenRouter	3	49	49	97.44%	\$0.03902909	4451.62 ms	3814.85 ms
OpenRouter	4	50	50	96.28%	\$0.04108734	3945.99 ms	3814.66 ms

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	OpenRouter leads overall (3/5 metrics)	Batch generation, offline summaries, backend processing	First-token and E2E response are faster; check cache and cost
Price First	Minimize request and token cost	OpenRouter leads overall (3/5 metrics)	High-frequency templates, support automation, marketing, RAG prefixes	First-token and E2E response are faster; check cache and cost
Latency First	Minimize full-response waiting time	Infron leads overall (3/5 metrics)	Online chat, agent chains, IDE/writing assistants, realtime tools	Cache and cost are stronger; still check speed SLA
TTFT First	Minimize streaming first-token time	OpenRouter leads overall (3/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	First-token and E2E response are faster; check cache and cost

11. Conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	OpenRouter	Infron	Infron	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	OpenRouter	Infron	Infron	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Latency First	Minimize full-response waiting time	Infron	Infron	Infron	OpenRouter	OpenRouter	Infron leads overall (3/5 metrics)
TTFT First	Minimize streaming first-token time	OpenRouter	Infron	Infron	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short-window stability but not day-level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built-in templates do not cover every workload distribution	Use sanitized production-stratified corpora	Discuss representative long-context templates only

Limitation	Impact	Next step	Current handling
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	export/gpt54_nano_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions summary.json
Paired dataset	export/gpt54_nano_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_pairs.csv
Request-level dataset	export/gpt54_nano_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_requests.jsonl
Filtered structured records	export/gpt54_nano_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records.json
Excluded-record audit	export/gpt54_nano_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records_excluded.json
Test source	tests/test_rerun_routing_sort_cache_cost_ab.py
Benchmark runner source	scripts/rerun_routing_sort_cache_cost_ab.py
HTML report renderer source	scripts/render_glm52_deepseek_style_report.py
Dataset reference	business_representative built-in representative business templates; request-level export is benchmark_requests